

Infosys Springboard Virtual Internship 7.0 Completion Report

Team Details

Batch Number: 2

Team Number:1

Start Date: 15-07-2026

Internship Duration: 8 Weeks

Name: Syed Humera Tasleem

Project Deployment: <https://sitm-deploy.vercel.app>

Team GitHub Repository: https://github.com/ananyaravikumar548/Infosys_Support-AI-Ticket-Management-.git

Personal GitHub Repository:<https://github.com/humeratasleem-sd/SupportAITicketManagementAgent>

1. Project Title

SupportPilot – AI-Powered Customer Support and Ticket Resolution Platform

2. Project Objective

The main objective of the **SupportPilot internship project** was to design and develop an AI-powered support ticket management platform that improves the **speed, efficiency, consistency, and overall quality of technical support operations**.

The project aimed to provide an end-to-end solution for managing support requests, starting from **ticket creation and classification** through **priority assessment, knowledge retrieval, AI-assisted resolution, escalation, customer communication, and ticket closure**. The system was designed to reduce repetitive manual effort for support teams while helping them make faster and more informed decisions.

A key objective was to use AI responsibly in the support workflow. Simple and high-confidence issues could be supported through automated resolutions, while complex, critical, or low-confidence issues could be **escalated to human support agents with the relevant ticket analysis and context already available**. This approach helps maintain human oversight while improving operational efficiency.

The project is relevant to organizations that handle a high volume of IT support requests because it can help **reduce response and resolution time, improve ticket prioritization, increase support-team productivity, provide better visibility into ticket operations, and deliver a more structured customer-support experience.**

The completed project was deployed as a web application and maintained through a team GitHub repository.

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3. Project description in detail

SupportPilot was developed as an end-to-end, role-based support ticket management platform designed to streamline the complete lifecycle of technical support requests. The system provides dedicated workflows for **Customers, Support Agents, Support Managers, and Administrators**, with each role receiving appropriate features and access based on its responsibilities. The development was structured into **four milestones across eight weeks**, allowing the project to progress systematically from core ticket management to an integrated AI-assisted support solution.

Milestone 1 – Ticket Management Foundation (Weeks 1–2)

The first milestone established the core infrastructure of the SupportPilot platform. The implementation included **user authentication, JWT-based security, role-based access control, ticket creation and management, ticket queues, ticket details, status transitions, dashboards, REST APIs, and database integration.**

The system also introduced **ticket classification support, priority and severity handling, and SLA-related information.** Customers could create tickets and monitor their progress, while support agents could access queues and manage ticket status. Managers and administrators were provided with broader operational visibility.

Milestone 2 – AI Intelligence and Knowledge Retrieval (Weeks 3–4)

The second milestone introduced the AI capabilities of the platform. The system was enhanced to understand incoming tickets and support intelligent decision-making through **classification, priority analysis, sentiment analysis, similar-ticket detection, knowledge retrieval, and Retrieval-Augmented Generation (RAG).**

The RAG workflow retrieves relevant information from trusted support documentation, troubleshooting guides, and knowledge sources before generating a response. This enables the system to produce **context-aware and knowledge-grounded troubleshooting recommendations** rather than relying only on general AI-generated information. The milestone also established the foundation for AI-assisted resolution generation.

Milestone 3 – Multi-Agent Resolution and Enterprise Integration (Weeks 5–6)

The third milestone extended the AI capabilities into a coordinated **multi-agent workflow.** An AI Orchestrator manages specialized agents responsible for **diagnosis, knowledge retrieval, resolution generation, validation, and escalation.**

The workflow first analyses the issue and retrieves relevant knowledge. A resolution is then generated and validated through a confidence check. **High-confidence cases can proceed toward automated resolution**, while **low-confidence, complex, or unsuitable cases are escalated to human support**.

This milestone also introduced **Jira integration, automated email communication, workflow tracking, activity logging, and resolution monitoring**, allowing the AI workflow to connect with enterprise support operations.

Milestone 4 – Dashboard, Optimization, Testing and Deployment (Weeks 7–8)

The fourth milestone focused on completing the operational side of the platform. The project was enhanced with **support analytics, dashboard capabilities, monitoring, workflow optimization, testing, and deployment preparation**.

The final stage emphasized validating the complete ticket lifecycle, improving the application's performance and usability, and ensuring that the integrated workflow could operate from ticket creation through resolution or escalation. The project blueprint defines this stage around dashboard development, optimization, final evaluation, performance improvement, and production deployment.

Technology Used

The project was developed using a combination of modern web-development, backend, AI, database, integration, and deployment technologies:

Frontend: React.js, Tailwind CSS, React Router, Axios, Recharts

Backend: Python, Django, Django REST Framework

Authentication: JWT / Simple JWT

Database: MongoDB Atlas

AI Technologies: AI/ML services, RAG, semantic knowledge retrieval, multi-agent workflows

Enterprise Integration: Jira, Email Automation

Development & Version Control: Git, GitHub

Deployment: Vercel and Render

Real-World Relevance

The real-world value of SupportPilot lies in its ability to combine **traditional support ticket management with AI-assisted decision-making and automation**. The platform can handle suitable repetitive requests more efficiently while maintaining **human involvement for uncertain, critical, or complex cases**. This approach is intended to reduce repetitive workload, improve response and resolution efficiency, strengthen operational visibility, and provide a more structured support experience. The project specification similarly positions the platform as an AI layer between ticket intake and human support, with confident cases automated and other cases routed to the appropriate support team.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project onboarding, architecture understanding, repository setup and UI planning.	Set up the project structure, understood user roles and prepared the initial customer, agent and admin interfaces.
Week 2	Core ticket processing, authentication and ticket lifecycle.	Implemented ticket creation, ticket queues, role-based access, dashboards, ticket status flow and backend integration.
Week 3	AI ticket understanding and knowledge-base integration.	Integrated classification, priority/severity intelligence and knowledge retrieval workflows.
Week 4	RAG and AI resolution generation.	Implemented retrieval-based context, resolution suggestions, confidence-aware responses and supporting UI.
Week 5	Multi-agent workflow development.	Integrated orchestrator, diagnosis, retrieval, resolution, validation and escalation workflow screens and logic.
Week 6	Jira and email automation.	Integrated enterprise workflow visibility, Jira-related ticket activity, email communication and workflow logging.
Week 7	Dashboard, analytics and optimization.	Built/updated operational dashboards, ticket monitoring, assignment views and performance-oriented UI improvements.
Week 8	Testing, final integration and deployment.	Performed end-to-end validation, fixed integration issues, completed final UI flows and deployed the application.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project orientation, problem understanding, team coordination, architecture discussion and milestone planning.	Week 1
Prototype/First Draft	Initial customer/agent/admin interfaces, authentication, ticket creation and core ticket-management flow.	Week 2
Mid-Term Review	Demonstration of AI classification, RAG/knowledge retrieval, resolution flow and multi-role integration.	Week 4–5
Final Submission	M1–M4 features integrated, tested and deployment completed.	Week 8
Presentation	Final project demonstration covering user roles, AI workflow, escalation, dashboards and deployment.	Week 8

5b. Project execution details

The SupportPilot project was executed using a **structured milestone-based development approach** over eight weeks. The development strategy focused on building and validating the core support-ticket infrastructure first and then progressively integrating AI, knowledge retrieval, multi-agent workflows,

enterprise integrations, analytics and deployment capabilities. This approach helped the team manage the complexity of the project while ensuring that every milestone provided a stable foundation for the next stage.

Initial Planning and System Foundation

The project began with analysis of the support-domain requirements, identification of user roles, definition of ticket states, and planning of the overall application workflow. The system was designed around four main stakeholders: **Customer, Support Agent, Support Manager and Admin.**

The team first established the core ticket lifecycle and API structure so that customer actions, agent activities and administrative operations could be handled consistently. This foundation was important because the later AI services depend on reliable ticket information, user identity, status history and operational data. The technical blueprint similarly separates the transactional support layer from the AI layer so that AI processing can work on a dependable system of record.

Milestone 1 – Core Ticket Management

During M1, the primary focus was on developing the operational foundation of the platform. Authentication and authorization were implemented using JWT-based security, followed by role-based access controls for different users.

The ticket-management workflow was then developed, including:

- Customer ticket creation
- Ticket validation
- Ticket listing
- Ticket detail views
- Ticket queues
- Ticket status management
- Priority and severity information
- SLA-related information
- Backend REST APIs
- Database connectivity
- Customer and agent dashboards

The frontend was structured so that customers could independently create and track their tickets while support users received dedicated interfaces for managing support operations.

The agent workflow included filtering, searching and reviewing tickets, while the customer workflow focused on ticket submission and progress tracking. The project specification also included classification support and the ability for support personnel to correct AI classification results so that those corrections could later contribute to improving the system.

Milestone 2 – AI Intelligence and Knowledge Retrieval

After the ticket-management foundation was established, M2 introduced the AI capabilities.

The first stage of the AI workflow focused on understanding the ticket itself. Ticket information was analysed to identify relevant categories, sub-categories and priority indicators. Additional intelligence

such as **sentiment analysis and similar-ticket detection** was incorporated to improve the understanding of the support request.

The next stage introduced **Retrieval-Augmented Generation (RAG)**. Instead of generating a response directly, the system first searches trusted knowledge sources such as support articles, troubleshooting guides and other relevant documentation. The retrieved information is then passed as context for generating a support recommendation.

This approach creates a more grounded resolution workflow because the AI response is based on the retrieved support context. The project documentation specifically defines M2 around knowledge retrieval, RAG workflows and automated troubleshooting responses.

The frontend was extended to represent this information through AI-related panels, knowledge suggestions, resolution displays and supporting context where required.

Milestone 3 – Multi-Agent Workflow and Enterprise Integration

In M3, the AI workflow was expanded from individual AI capabilities into a coordinated multi-agent architecture.

Instead of treating the entire resolution process as a single operation, the workflow was divided into specialized stages:

Ticket → Orchestrator → Diagnosis → Knowledge Retrieval → Resolution Generation → Validation → Escalation / Resolution

The **Orchestrator** manages the workflow and maintains the sequence of operations. The **Diagnosis Agent** identifies possible causes and important details about the issue. The **Knowledge Retrieval Agent** obtains relevant information from the existing M2 knowledge system. The **Resolution Agent** generates a step-by-step response using the ticket, diagnosis and retrieved evidence. Finally, the **Validation stage** checks whether the generated resolution is sufficiently relevant and reliable.

A confidence-based decision was then introduced. High-confidence issues could proceed toward automated resolution, while low-confidence or complex cases could be transferred to human support.

M3 also connected the AI workflow with enterprise support processes through **Jira integration, email automation and activity logging**. This ensured that escalation and resolution activities could remain synchronized with the overall ticket lifecycle.

On the frontend, additional operational screens were introduced for AI workflow status, agent execution information, confidence levels, Jira information, email history and ticket activity.

Milestone 4 – Operational Readiness, Analytics and Deployment

The final milestone focused on converting the developed modules into a more complete operational system.

The work included completion and refinement of:

- Analytics dashboards
- Manager views
- Ticket monitoring
- Escalation visibility
- Performance-related improvements

- AI quality information
- Testing and validation
- Deployment preparation

Testing was carried out across different ticket states and workflow conditions, including normal tickets, AI-resolved tickets, escalated tickets, customer confirmation, reopening scenarios and role-based access.

The project blueprint defines M4 around analytics, performance optimization, final evaluation and deployment readiness, including query optimization, caching, latency analysis and final testing across the system.

Frontend Development and My Contribution

My primary contribution to the project was **frontend development**. I worked on converting the functional and technical requirements into usable interfaces for different roles.

My frontend contribution included the development and integration of:

- Customer dashboard
- Customer ticket creation page
- My Tickets page
- Ticket status and tracking views
- Ticket detail screens
- Agent ticket queue
- Agent ticket details
- Manager console
- Assignment and filtering interfaces
- Dashboard cards and operational views
- AI resolution presentation
- Status and priority indicators
- Workflow-related components
- API-connected frontend services

I also worked on maintaining a consistent design and navigation experience across different roles. The frontend had to display information from multiple parts of the backend and AI workflow without exposing internal technical information unnecessarily to customers.

The M1 frontend specification includes dedicated ticket creation, ticket listing, ticket detail, queue and classification interfaces, while later milestones extend the frontend with AI operations and integration screens.

API and Frontend Integration

An important part of execution was connecting the frontend with the backend REST APIs. The frontend consumed APIs for authentication, ticket creation, ticket retrieval, status updates, assignment-related operations and AI workflow information.

This required maintaining consistent request and response structures and ensuring that changes in ticket state were reflected correctly in the user interface.

The development process therefore involved repeated cycles of:

UI Development → API Integration → Testing → Debugging → Refinement

This helped identify issues in data handling, state changes, role-based visibility and workflow presentation before final deployment.

Testing and Validation

Testing was performed throughout the development process instead of being limited to the final stage.

The project included validation of:

- Customer ticket creation
- Required-field validation
- Ticket classification
- Priority handling
- AI confidence conditions
- Knowledge retrieval
- AI-generated resolutions
- Human escalation
- SLA-related conditions
- Ticket status transitions
- Role-based authorization
- Jira workflows
- Email communication
- Customer confirmation
- Ticket reopening and closure

The project architecture also defines fallback handling so that when the AI cannot safely resolve an issue, the ticket is transferred to human support rather than being treated as successfully resolved.

Deployment

After completing the major development and testing activities, the application was prepared for deployment. Git and GitHub were used for version control and collaborative development, while the project frontend was deployed through Vercel and the backend deployment was supported through the project deployment setup.

Live Deployment: <https://sitm-deploy.vercel.app>

Team GitHub Repository: https://github.com/ananyaravikumar548/Infosys_Support-AI-Ticket-Management-.git

Overall Execution Approach

The overall technical approach was based on using AI as an **assistive layer rather than completely replacing human support**.

The final workflow can be summarized as:

Customer Ticket → AI Analysis → Knowledge Retrieval → Resolution Generation → Confidence Validation → Automated Resolution OR Human Escalation

This design makes the system more practical for enterprise support because routine and high-confidence issues can be handled efficiently, while uncertain, critical or complex issues can be transferred to human agents with the AI analysis and relevant context already available. The project architecture explicitly follows this confidence-based automation and human-fallback model.

6. Snapshots / Screenshots

The following screenshots provide visual evidence of the implemented customer, manager and agent workflows.

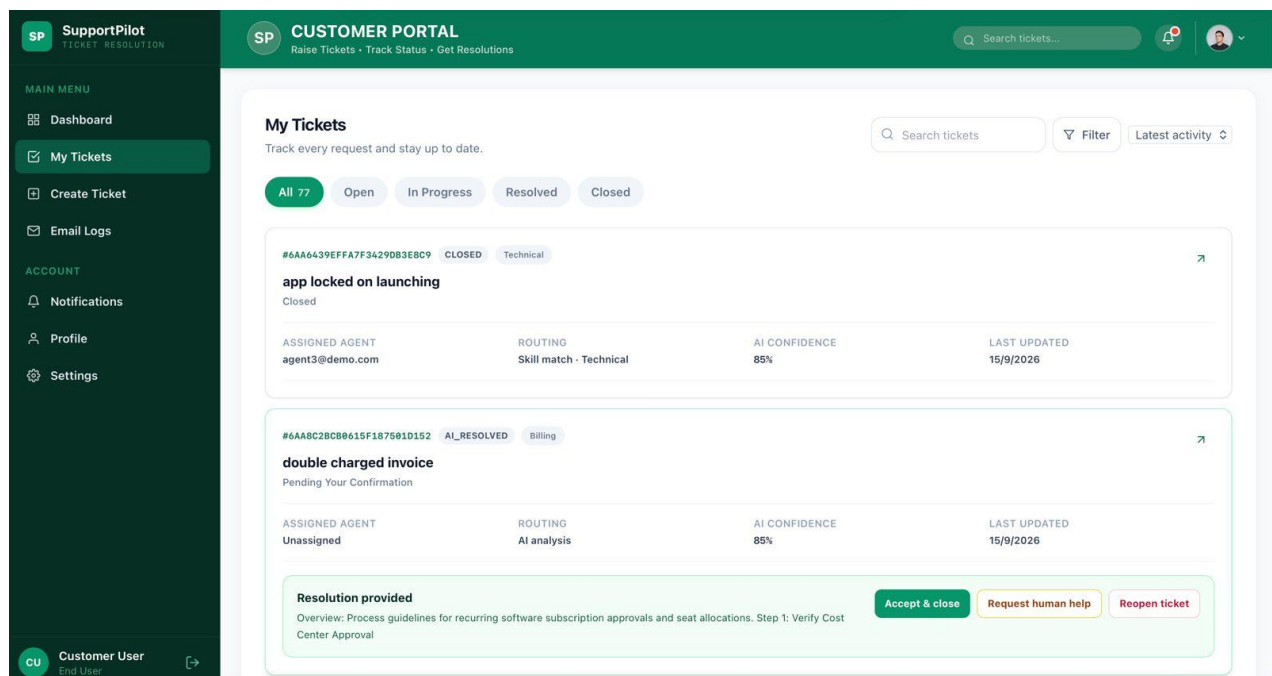


Figure 1. SupportPilot Customer My Tickets – ticket status, routing information, AI confidence and resolution actions.

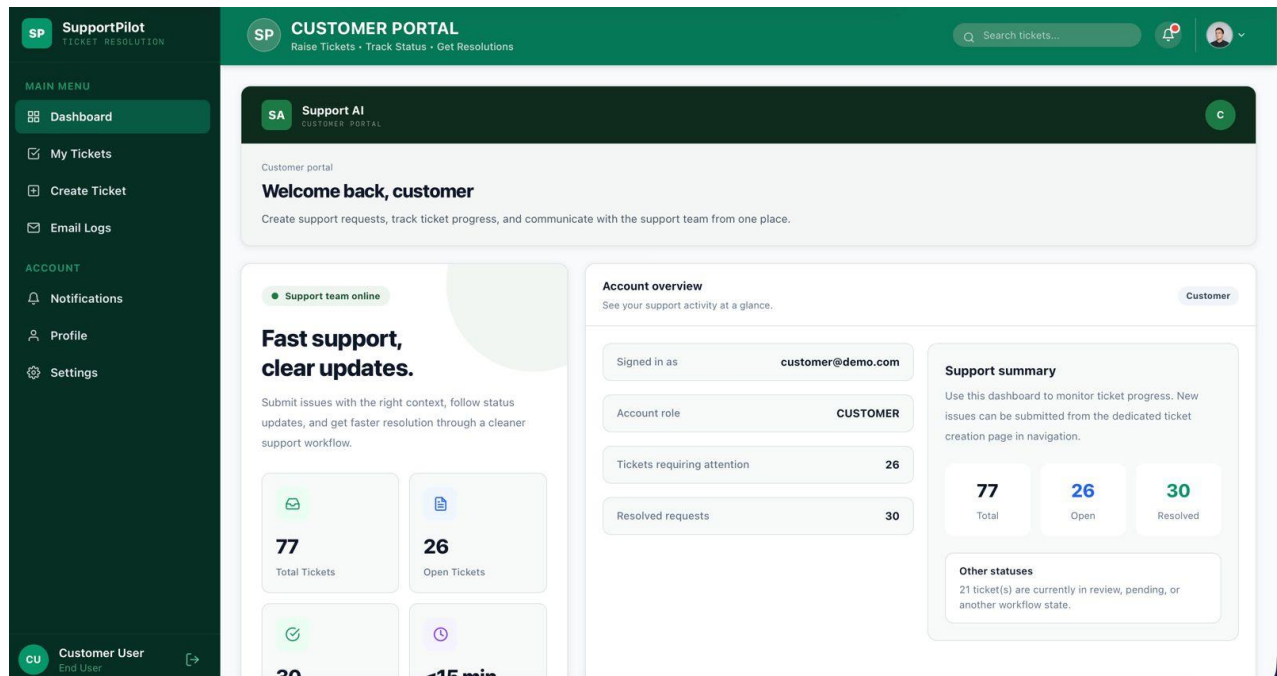


Figure 2. SupportPilot Customer Dashboard – ticket summary, support activity and account overview.

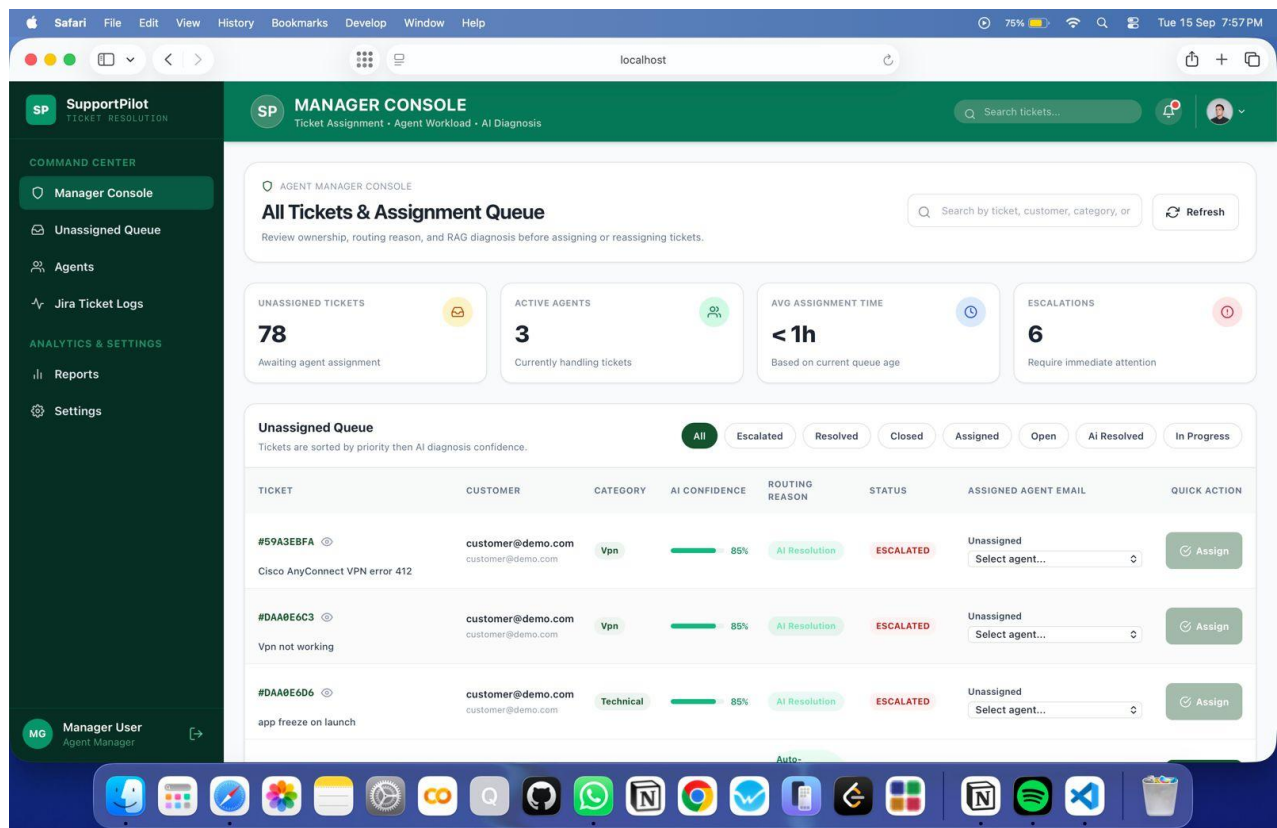


Figure 3. SupportPilot Manager Console – unassigned queue, active agents, escalations and ticket assignment.

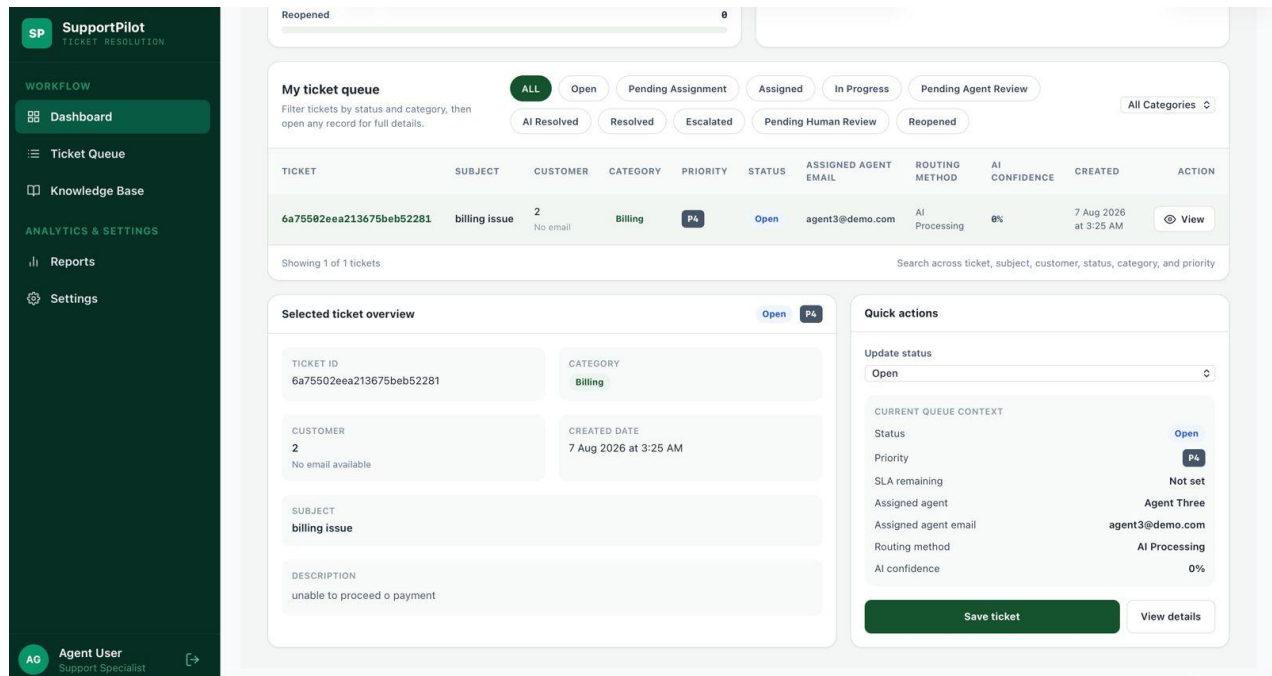


Figure 4. SupportPilot Agent Console – ticket queue, priority, status, assigned agent, SLA information and AI processing.

7. Challenges Faced

Several technical, implementation and coordination challenges were encountered during the development of the SupportPilot project. These challenges provided practical experience in integrating multiple technologies and maintaining a consistent workflow across the application.

1. Frontend and Backend Integration

Different frontend screens depended on backend APIs for authentication, ticket data, status updates, assignments and AI-related information. Changes in API response structures or data requirements could affect multiple screens at the same time.

This was addressed by maintaining consistent API usage, validating response data before rendering it in the UI, handling loading and error states, and testing complete user workflows rather than testing screens in isolation.

2. Role-Based Application Complexity

SupportPilot contains separate workflows for **Customer, Support Agent, Support Manager and Admin** users. Each role has different permissions, responsibilities and information requirements.

Managing these differences required careful handling of route protection, role-based navigation and UI visibility. Separate role-based layouts and access controls were used to keep the workflows organized and prevent users from accessing functionality outside their responsibilities.

3. AI Workflow Integration

The AI workflow consisted of multiple connected stages, including **classification, priority and sentiment analysis, similar-ticket detection, knowledge retrieval, resolution generation, confidence validation and escalation**.

Integrating these components into a single ticket lifecycle was challenging because a problem in one stage could affect the subsequent stages. The workflow was therefore divided into logical processing stages, making individual components easier to test, debug and integrate.

4. Dynamic Ticket States and Workflow Handling

Tickets could move through several states such as **Open, In Progress, AI Resolved, Escalated, Resolved, Closed and Reopened**.

Displaying these states consistently across customer, agent and manager interfaces required careful UI handling. Status badges, filters, workflow actions and ticket details had to remain synchronized with the actual ticket state.

The project architecture defines separate transitions for normal resolution, escalation and reopening, which helped provide a structured approach to this challenge.

5. Deployment and Environment Configuration

Connecting the frontend application with backend services and managing environment-specific configurations required additional testing. Differences between local development and deployed environments could affect API connectivity, authentication and application behaviour.

The team addressed these issues through environment-aware configuration, repeated integration testing and deployment verification before finalizing the application.

6. AI and Human Workflow Coordination

Another important challenge was determining when an issue should be handled automatically and when it should be transferred to human support.

The project adopted a confidence-based approach. High-confidence issues could proceed toward automated resolution, whereas low-confidence or complex issues were routed to human agents. This helped maintain human oversight while still providing the efficiency benefits of AI-assisted support.

7. Enterprise Integration Complexity

The integration of **Jira, email automation and workflow logging** introduced additional dependencies. Ticket information, resolution status and communication events needed to remain synchronized across the support platform and external services.

Testing these integrations and handling different workflow outcomes required careful coordination between the ticketing, AI and integration components. The M3 project requirements specifically included Jira synchronization, email events and complete activity logging.

8. Team Coordination

Since the project was developed collaboratively, frontend, backend, AI and integration activities had to progress together. A change in one component could affect another team's implementation.

Regular communication, milestone-based reviews, shared API expectations and continuous integration of completed work helped reduce these coordination issues. The technical blueprint also emphasizes

written cross-team contracts and regular milestone reviews to avoid different teams working with inconsistent assumptions.

Overall Challenge Resolution

These challenges were addressed through **incremental development, continuous testing, structured milestone planning, clear separation of responsibilities and regular integration of team members' work**. The experience helped the team understand that successful software development requires not only coding individual features but also ensuring that all components work together as one reliable system.

8. Learnings & Skills Acquired

The internship provided valuable practical experience in developing an **AI-enabled enterprise support application**. Working on the SupportPilot project helped me strengthen my technical knowledge, understand AI-based support workflows, and develop important professional skills through a structured, milestone-based development process.

Technical Skills Acquired

I gained practical experience in:

- **React.js** for developing interactive and role-based web interfaces.
- **Tailwind CSS** for creating responsive and consistent user interfaces.
- **React Router** for navigation and role-based routing.
- **Axios and REST APIs** for connecting frontend applications with backend services.
- **Dashboard development** using reusable cards, tables, filters and data components.
- **Reusable component development** to maintain consistency across different screens.
- **Form handling and validation** for reliable ticket submission and user interaction.
- **Role-based UI development** for Customer, Support Agent, Support Manager and Admin workflows.
- **Git and GitHub** for version control and collaborative development.
- **Deployment and environment configuration** for making the application available in a hosted environment.
- **Debugging and integration testing** to identify and resolve frontend and API-related issues.

AI and Domain Knowledge Acquired

The project provided practical exposure to AI concepts used in technical support systems, including:

- Ticket classification and categorization
- Priority and severity analysis
- Sentiment analysis
- Similar-ticket detection
- Knowledge-base retrieval
- Retrieval-Augmented Generation (RAG)

- AI-assisted resolution generation
- Confidence-based resolution and escalation
- Multi-agent AI workflows
- Human escalation and support routing
- SLA and ticket lifecycle management

I also developed an understanding of how different AI stages can work together as a single workflow, from **ticket analysis and knowledge retrieval to resolution generation, validation and escalation**.

Project Development Skills Acquired

The internship strengthened my ability to work through a complete software-development lifecycle. I gained experience in:

- Understanding project and functional requirements
- Planning work according to project milestones
- Designing user workflows based on different roles
- Integrating frontend interfaces with backend APIs
- Testing complete application workflows
- Identifying and resolving defects
- Handling changes during development
- Supporting deployment activities
- Preparing project demonstrations and technical documentation

Professional Skills Acquired

The internship also helped me develop important professional skills, including:

- **Team Collaboration:** Working closely with frontend, backend and AI team members.
- **Communication:** Discussing requirements, technical issues and integration-related changes with the team.
- **Task Ownership:** Taking responsibility for assigned frontend activities and completing them within the planned milestone.
- **Problem Solving:** Analysing issues, debugging errors and implementing practical solutions.
- **Time Management:** Completing assigned activities within the defined eight-week development schedule.
- **Adaptability:** Learning and applying new technologies while working on an evolving project.
- **Documentation:** Understanding the importance of maintaining clear project and technical documentation.

Overall Learning

The internship helped me understand that developing an AI-powered enterprise application requires more than implementing AI functionality. A successful system also depends on **well-designed frontend interfaces, backend APIs, authentication, database integration, workflow management, testing, deployment and effective teamwork**.

Overall, the internship helped bridge the gap between academic learning and practical software development. It improved my confidence in frontend development, strengthened my understanding of AI-assisted enterprise applications, and provided valuable experience in working as part of a professional development team.

9. Testimonials from team

The internship provided a valuable opportunity to work as part of a collaborative development team and understand how different technical responsibilities contribute to the development of one complete product.

The project experience strengthened my confidence in **frontend development, API integration, debugging and technical presentation**. Working closely with different parts of the system helped me understand how frontend, backend and AI components need to work together to deliver a reliable application.

Following a milestone-based development process also improved my ability to **prioritize tasks, manage deadlines, troubleshoot technical issues and complete assigned features within a planned schedule**. The experience helped me become more comfortable with team-based development and gave me greater confidence in contributing to real-world software projects.

Overall, the internship was a meaningful learning experience that improved both my **technical capabilities and professional approach to teamwork and software development**.

10. Conclusion

The Infosys Springboard Virtual Internship was a valuable practical learning experience that provided exposure to the development of an AI-enabled enterprise application. Through the **SupportPilot** project, the team developed an integrated support workflow covering ticket creation, AI-based analysis, knowledge retrieval, resolution generation, confidence-based automation, human escalation, enterprise integration, analytics and deployment.

The project was completed through a structured four-milestone development approach, which helped transform the initial ticket-management foundation into a more comprehensive AI-assisted support platform. Each milestone added a new layer of functionality while building upon the previous stage, resulting in a connected end-to-end support workflow.

My primary contribution was in frontend development, where I worked on customer-facing, support-agent and manager-facing interfaces, along with API integration and workflow-related components. This provided practical experience in developing user interfaces that communicate with different parts of an integrated enterprise application.

The internship strengthened my technical knowledge as well as my ability to understand requirements, work collaboratively, integrate different modules, troubleshoot issues, manage milestone-based tasks and contribute to the delivery of a working application.

Overall, the experience helped bridge the gap between academic knowledge and practical software development. It has strengthened my interest and confidence in software development, frontend engineering and AI-enabled applications, making the internship highly relevant to my future academic and career goals.

11. Acknowledgements

I sincerely thank **Infosys Springboard** for providing me with the opportunity to participate in the Virtual Internship Program. I am grateful to the mentors and coordinators for their continuous guidance, valuable feedback, reviews and support throughout the internship.

I would also like to thank my **Team 1 teammates** for their collaboration, technical discussions and support during the development, integration and testing of the **SupportPilot** project. Working as part of a team helped me understand the importance of communication, coordination and shared responsibility in completing a real-world software project.

This internship provided meaningful exposure to practical software-development practices and helped me strengthen my **technical knowledge, problem-solving ability, teamwork, communication and professional skills**.

Project Links

Live Application:

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